

Module A (opener)

Fama–French factors on a 50-name panel

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Paper card: Fama & French (2015)

	<i>Journal of Financial Economics</i> 116: 1–22
Question	Do profitability (RMW) and investment (CMA) factors, added to the three-factor model, explain average stock returns better?
Data	US stocks, July 1963 – December 2013 (606 months); portfolios sorted on size, B/M, profitability and investment (25 or 32 per sort).
Method	Time-series regressions of portfolio excess returns on the factors.
How it is tested	Intercepts, not R^2 : the GRS test, the average absolute intercept, and the share of the cross-section of expected returns left unexplained.
Main result	The five-factor model beats FF3 and explains 71–94% of the cross-section variance of expected returns, yet GRS rejects every model. HML becomes redundant. Main failure: small stocks that invest a lot despite low profitability.
Watch for	The model is judged on average returns (intercepts), not on explained variance. In our notebook: does a 50-stock daily panel have the power to reject?
Code & data	Factors and portfolios are free in Ken French's data library.

Fama–French as a factor model

Fama, E. F., & French, K. R. (1993). *Common risk factors in the returns on stocks and bonds*. *JFE*, 33(1), 3–56.

Fama, E. F., & French, K. R. (2015). *A five-factor asset pricing model*. *JFE*, 116(1), 1–22.

This uses factor returns, not firm-level book-to-market.

Live: `notebooks/01_ff_factors_real.ipynb`

Data: `krauss_sp500_daily.csv` + `ff_daily_factors.csv`

One factor path, 50 betas

$$r_{i,t} - r_{f,t} = \alpha_i + \beta'_i \mathbf{FF}_t + \varepsilon_{i,t}$$

Same for all stocks	Stock-specific
Mkt-RF, SMB, HML, ...	α_i, β_i

Ken French library → one row per **day**.

Not the Gu target

01a / 01b	01
Predict r_{t+1} from characteristics	Explain r_t with today's factors
Predictive OOS R^2	Explanatory R^2 (betas frozen)

Do not mix the two R^2 .

How FF judge a model: intercepts, not R^2

R^2 measures variance explained. An asset pricing model is judged on **average returns** explained, so FF (2015) test whether all $\alpha_i = 0$:

$$\text{GRS} = \frac{T - N - L}{N} \frac{\hat{\alpha}' \hat{\Sigma}^{-1} \hat{\alpha}}{1 + \bar{\mu}' \hat{\Omega}^{-1} \bar{\mu}} \sim F(N, T - N - L)$$

Report GRS **and** mean $|\hat{\alpha}_i|$. In our notebook GRS does **not** reject (FF3: $p \approx 0.05$ in 2015–19, 0.28 in 2020–23), yet mean $|\hat{\alpha}_i| \approx 8\%$ a year. Daily returns are noisy: large alphas can be jointly insignificant.

Frozen protocol (still)

- Calendar train / test split (e.g. through 2019 / from 2020)
- Estimate β on the training period only
- Test: $\widehat{r - r_f} = \hat{\beta}'_{\text{train}} \text{FF}_t$
- FF3 vs FF5: compare **GRS** and mean $|\alpha|$ in the **test** period, not only R^2

Who are the 50 names?

Today's large S&P 500 firms, downloaded from Yahoo.

We picked them **because they survived and grew**, so their $\hat{\alpha}_i$ are biased upward.

This has nothing to do with the factor model. It is **survivorship**, and we meet it again in Krauss.

Then the ML (same notebook, §8)

The features are known **yesterday**: lagged FF factors + own return lags.

One pooled model: OLS $\rightarrow$ Ridge $\rightarrow$ RF.

Predictive R^2 vs a forecast of 0: ≈ 0 .

Contemporaneous FF R^2 : ≈ 0.45 .

A high R^2 with today's factors is **not** a forecast.

Takeaway

FF OLS is attribution, judged by intercepts. Ridge/RF on lags is the ML forecast.

Gu's next move is a **characteristic zoo** and a monthly horizon, not a deeper fit of today's SMB.

Next: 01a (Gu)